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Security

“Artificial Intelligence, Deep Learning And Machine Vision Are... Heralding A NEW AGE Of SECURITY SOLUTIONS”

As scale and complexity of businesses increase, enterprises will have to move towards smarter, centralised and automated surveillance, access control and attendance management solutions. But how are these devices becoming smarter and technically-advanced? Ganesh Jivani, managing director, Matrix Comsec, explains to Paromik Chakraborty of *Electronics For You*

Q. What latest technologies are getting embedded with surveillance systems?

A. For video surveillance, organisations are moving towards Internet Protocol (IP) by replacing their analogue solutions. Analogue DVRs are being replaced by IP network video recorders. Image resolution is increasing from 2MP to 4MP, and beyond. Video codec H.265 is also becoming the norm. Video is storage-hungry, so hard-disks now come with increased storage capacities.

Interestingly, an increasing number of multi-location organisations are investing in video management systems to integrate and centralise their existing isolated legacy solutions.

Video analytics is another exciting area, as analytics are becoming more reliable and useful for real-life applications. Analytics such as face recognition are erasing the boundary between video surveillance and biometric technologies.

Users are moving from desktop and laptop to mobile as their choice of device for video surveillance, fuelling the need for innovative mobile apps.

We allow complete flexibility to customers to create real-life scenarios by using intelligent video analytics and integrations to automate identification as well as notifications based on scenarios.

Q. How will this impact the upfront cost and opex for surveillance equipment? How long will it take to recover these costs?

A. Even though IP video surveillance is somewhat costly when compared to analogue CCTVs, in return, users benefit from new-generation

functions, features, performance and longer product lifecycles. But, IP video surveillance is fast approaching the price range of analogue solutions.

The ROI differs based on industry, application, solution design and business utility. But at a broad level, we can expect an ROI in two years.

Q. What are the most important features in surveillance equipment?

A. In addition to IP video surveillance, attention should be paid to resolution, video quality in different usage scenarios, streaming, bandwidth and storage consumption, and video analytics. In case of two or more locations, users should evaluate investing in a software-centric, centralised video surveillance solution for real benefits.

Q. What crucial market challenges should customers be aware of?

A. They need to be wary of product specifications, and insist on proof-of-concept in real-life scenarios.

Then, instead of focussing only on upfront product prices, they should evaluate total cost of ownership, which includes purchase cost, peripherals, product life, service cost, impact of breakdown on business and replacement cost, and compliance cost.

Finally, they should explore investing in software-centric solutions for flexibility and intelligent functions and features of automation.

Q. How are biometric access systems improving?

A. Organisations are increasingly moving to biometric solutions based

on finger, palm, face or iris scans. They are demanding innovative solutions on integration with ERP, CRM, Google Maps, video surveillance and building management systems. Cloud-based SaaS model is also gaining popularity in security solutions.

Organisations are shifting from scattered and localised monitoring to centralised monitoring, which helps synchronise the organisations and gives higher control to management besides providing enhanced data security. These also improve processes and increase efficiency manifold.

Q. How important is cloud technology for security solutions?

A. Cloud is gaining ground rapidly due to obvious customer benefits. Bandwidth availability is increasing and cost is reducing, spurring cloud adoption. Video surveillance and biometrics are transforming to software-centric solutions, further accelerating the need to host these applications on the cloud.

Q. What are India's strengths and weaknesses in this domain? Any scope left for innovation?

A. Our strengths include manpower, cost control and large local markets. And limitations include having fewer product companies and lack of knowledge of global customer requirements.

Video surveillance and access control have huge scope for innovation. In particular, artificial intelligence, deep learning and machine vision are emerging areas heralding a new age of security solutions. **EFY**